



Killara High School Extension 2 Mathematics

Year 12
JUNE 2011

TOTAL TIME: 60 minutes

TOTAL MARKS: 40

Instructions

1. Show all necessary working.
2. Marks may be deducted for a lack of working or poor setting out of solutions.
3. Recommended time per question:
question 1, 34 minutes and
question 2, 26 minutes.

NAME

Question 1 (23 marks)**Marks**

a) Find $\int \frac{dx}{\sqrt{3-2x-x^2}}$ 2

b) Find $\int \frac{dx}{(x-2)(x-3)}$ 2

c) Find $\int 2x \ln(x^2+1) dx$ 2

d) Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{x^2 dx}{\sqrt{1-x^2}}$ 3

e) i. Prove that if $I_n = \int \tan^n \theta d\theta$, where n is a non-negative number, then $I_{n+2} = \frac{\tan^{n+1} \theta}{n+1} - I_n$ 3

ii. Hence find I_4 1

f) Evaluate $\int_1^2 \frac{25 dx}{(x+2)(2x-1)^2}$ 4

g) Evaluate $\int_0^{\frac{\pi}{4}} x \sin 2x dx$ 2

h) Use the 't' method to evaluate $\int_0^{\frac{\pi}{3}} \frac{\tan x}{1+\cos x} dx$. 4

End of Question 1

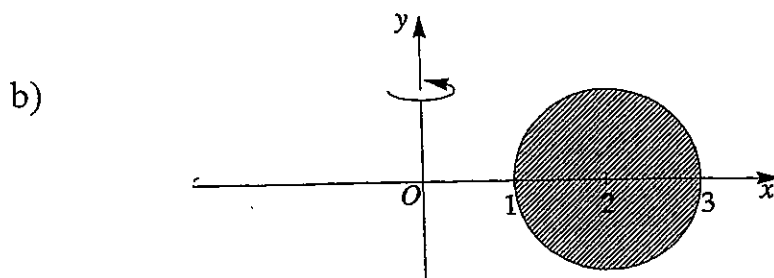
Question 2 (17 marks)**Marks**

- a) The base of a certain solid is an equilateral triangle of side length S , with one vertex at the origin and an altitude along the x -axis of length p units. Each plane cross-section perpendicular to the x -axis is a square of side length $2y$ units, one side of which lies in the base of the solid.

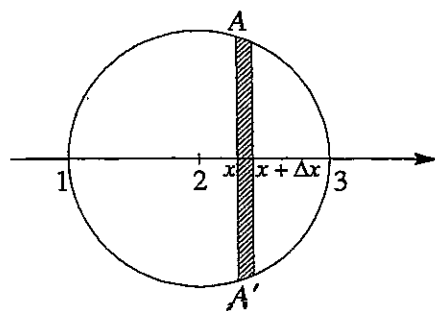
- i. Show that the volume V of the solid is given by

$$V = 4 \int_0^p y^2 dx \quad 3$$

- ii. Calculate the volume of the solid. 2



The circular region $(x-2)^2 + y^2 \leq 1$ is rotated about the y -axis.



- i. Show that $AA' = 2\sqrt{1-(x-2)^2}$. 2

- ii. Show that the volume ΔV obtained when a typical strip of height AA' and thickness Δx is rotated about the y -axis is given by

$$\Delta V = 4\pi x \sqrt{1-(x-2)^2} \Delta x. \quad 1$$

- iii. Find the total volume of the solid generated. 3

Question 2 continues over the page

Question 2 continued

- c) The region bounded by the curve $y = x^3$, the line $y = 1$ and the y-axis is rotated about the line $y = -1$.
- i. Show that if slices are taken perpendicular to the x-axis then the volume of a typical slice is given by:

$$\Delta V = \pi(3 - 2x^3 - x^6)dx \quad 3$$

- ii. Calculate the volume of the solid generated. 3

The End

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

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Lx 2

Solutions

$$\begin{aligned} a \int \frac{dx}{\sqrt{3-2x-x^2}} \\ &= \int \frac{dx}{\sqrt{4-(x+1)^2}} \quad \checkmark \\ &= \sin^{-1}\left(\frac{x+1}{2}\right) + C \quad \checkmark \end{aligned}$$

$$\begin{aligned} b \int \frac{dx}{(x-2)(x-3)} \\ &= \int \frac{-1}{x-2} dx + \int \frac{1}{x-3} dx \quad \checkmark \\ &= -\ln(x-2) + \ln(x-3) + C \\ &= \ln\left(\frac{x-3}{x-2}\right) + C \quad \checkmark \end{aligned}$$

$$\begin{aligned} d \int_0^{\frac{1}{\sqrt{2}}} \frac{x^2}{\sqrt{1-x^2}} dx \\ \boxed{\begin{array}{l} \text{let } x = \sin \theta \quad \therefore dx = \cos \theta d\theta \\ \text{when } x = \frac{1}{\sqrt{2}}, \quad \theta = \frac{\pi}{4} \\ \text{when } x = 0 \quad \theta = 0 \end{array}} \\ &= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta \cos \theta d\theta}{\sqrt{1-\sin^2 \theta}} \quad \checkmark \\ &= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta \cos \theta d\theta}{\cos \theta} \\ &= \int_0^{\frac{\pi}{4}} \sin^2 \theta d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 - \cos 2\theta) d\theta \quad \checkmark \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left[\left(\frac{\pi}{4} - \frac{1}{2} \right) - (0 - 0) \right] \\ &= \frac{\pi-2}{8} \quad \checkmark \end{aligned}$$

$$c \int 2x \ln(x^2+1) dx$$

$$\begin{aligned} &\int \tan^{-1} x dx \quad \text{see over} \\ &= \int 1 \cdot \tan^{-1} x dx \quad \text{let } u = 1 \\ &= x \tan^{-1} x \quad \text{let } u = \tan^{-1} x \\ &\quad - \int x \cdot \frac{1}{1+x^2} dx \quad u' = \frac{1}{1+x^2} \\ &= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx \\ &= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C \end{aligned}$$

$$\begin{aligned} e \int \tan^n \theta d\theta \\ &= \int \tan^{n+2} \theta \cdot \cot^2 \theta d\theta \\ &= \int \tan^{n+2} \theta (\csc^2 \theta - 1) d\theta \\ &= \int \tan^{n+2} \theta \csc^2 \theta d\theta - \int \tan^{n+2} \theta d\theta \\ &= \int \tan^n \theta \sec^2 \theta d\theta - I_{n+2} \quad \checkmark \\ I_n &= \frac{\tan^{n+1} \theta}{n+1} - I_{n+2} \\ I_{n+2} &= \frac{\tan^{n+1} \theta}{n+1} - I_n \quad \checkmark \end{aligned}$$

$$\int 2x \ln(x^2+1) dx$$

$$\begin{aligned} \text{let } u &= x^2+1 \\ \therefore du &= 2x dx \end{aligned}$$

$$\begin{aligned} &= \int \ln u du \\ &= \int 1 \cdot \ln u du \\ &= \int \frac{d(\ln u)}{du} \ln u du \end{aligned}$$

$$\begin{aligned} &= u \ln u - \int u \cdot \frac{1}{u} du \quad \checkmark \\ &= u \ln u - \int 1 du \\ &= u \ln u - \frac{u^2}{2} + C \\ &= (x^2+1) \ln(x^2+1) - \frac{(x^2+1)^2}{2} + C \quad \checkmark \\ &= (x^2+1) \ln(x^2+1) - x^2 + C \end{aligned}$$

$$\text{If } I_{n+2} = \frac{\tan^{n+2} \theta}{n+1} - I_n$$

$$\begin{aligned} \text{then } I_4 &= \frac{\tan^3 \theta}{3} - I_2 \\ &= \frac{\tan^3 \theta}{3} - \int \tan^2 \theta d\theta \\ &= \frac{\tan^3 \theta}{3} - \int (\sec^2 \theta - 1) d\theta \checkmark \\ &= \frac{\tan^3 \theta}{3} - \tan \theta + \theta + C \end{aligned}$$

$$\text{f} \quad \text{let } \frac{25}{(x+2)(2x-1)^2} = \frac{a}{x+2} + \frac{b}{2x-1} + \frac{c}{(2x-1)^2}$$

$$\therefore 25 = a(2x-1)^2 + b(x+2)(2x-1) + c(x+2) \checkmark$$

$$x = \frac{1}{2}, \quad 25 = \frac{5}{2}c, \quad c = 10$$

$$x = -2, \quad 25 = 25a, \quad a = 1$$

$$x = 0, \quad 25 = 1 - 2b + 2c, \quad b = -2$$

$$\therefore \frac{25}{(x+2)(2x-1)^2} = \frac{1}{x+2} + \frac{-2}{2x-1} + \frac{10}{(2x-1)^2} \checkmark$$

$$\therefore \int \frac{25 dx}{(x+2)(2x-1)^2} = \int \frac{1}{x+2} - \frac{2}{2x-1} + \frac{10}{(2x-1)^2} dx \checkmark$$

$$\begin{aligned} &= \left[\ln(x+2) - \ln(2x-1) - 5(2x-1)^{-1} \right]_1^2 \\ &= \left(\ln 4 - \ln 3 - \frac{5}{3} \right) - \left(\ln 1 - \ln(-1) - 5 \right) \\ &= \frac{10}{3} + 2 \ln\left(\frac{2}{3}\right) \checkmark \end{aligned}$$

$$\text{g. } \int x \sin 2x dx$$

$$\text{let } u = x, v' = \sin 2x$$

$$u' = 1, v = -\frac{1}{2} \cos 2x$$

$$= \left[(x) \left(-\frac{1}{2} \cos 2x \right) \right]_0^{\pi/4} - \int_0^{\pi/4} \left(-\frac{1}{2} \cos 2x \right) dx \checkmark$$

$$= \left(-\frac{\pi}{8} \cos \frac{\pi}{2} \right) - (0) + \frac{1}{2} \int_0^{\pi/4} \cos 2x dx$$

$$= 0 + \frac{1}{2} \times \frac{1}{2} \left[\sin 2x \right]_0^{\pi/4}$$

$$= \frac{1}{4} \left[\sin \frac{\pi}{2} - 0 \right] \checkmark$$

$$= \frac{1}{4}$$

$$\text{h. } \int_0^{\pi/3} \frac{\tan x}{1 + \cos x} dx$$

$$\text{let } t = \tan \frac{x}{2}$$

$$\therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$dx = \frac{2dt}{\sec^2 \frac{x}{2}}$$

$$dx = \frac{2dt}{1+t^2} \checkmark$$

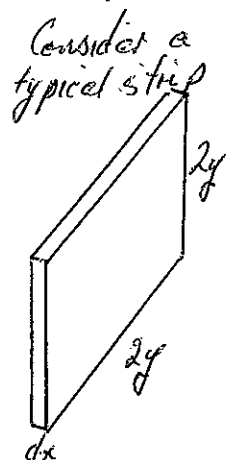
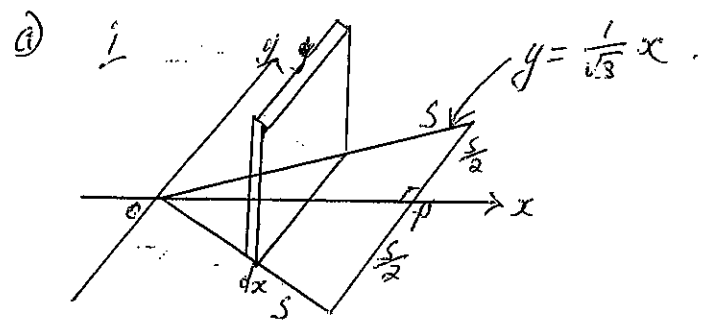
$$\int_0^{\pi/3} \frac{\tan x}{1 + \cos x} dx$$

$$= \int_0^{\pi/3} \frac{2t}{1-t^2} \times \frac{2dt}{1+t^2} \checkmark$$

$$= \int_0^{\pi/3} \left(\frac{4t}{1-t^2} \right) \div \left(\frac{2}{1+t^2} \right) dt$$

$$\begin{aligned} &= \int_0^{\pi/3} \frac{2t}{1-t^2} dt \checkmark \\ &= \left[-\ln(1-t^2) \right]_0^{\pi/3} \\ &= -\ln \frac{2}{3} + \ln 1 \\ &= -\ln \frac{2}{3} \\ &= \ln \frac{3}{2} \checkmark \end{aligned}$$

Question 2



$$dV \equiv 4y^2 dx$$

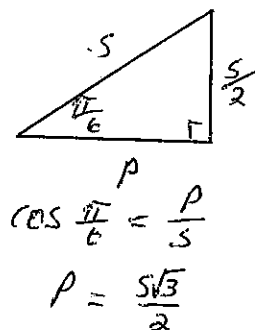
$$\begin{aligned} \text{Total Volume} &\equiv \int_{dx=0}^P 4y^2 dx \\ &= \lim_{dx \rightarrow 0} \sum_{x=0}^P 4y^2 dx \\ &= 4 \int_0^P y^2 dx \end{aligned}$$

ii

$$\text{Total Volume} = 4 \int_0^{\frac{S\sqrt{3}}{2}} \frac{1}{3} x^2 dx$$

$$\begin{aligned} &= \frac{4}{3} \left[\frac{x^3}{3} \right]_0^{\frac{S\sqrt{3}}{2}} \\ &= \frac{4}{3} \left[\frac{1}{3} \cdot \frac{S^3}{8} \cdot \frac{3\sqrt{3}}{8} - 0 \right] \\ &= \frac{S^3 \sqrt{3}}{6} \end{aligned}$$

Must be
just in terms
of S or P



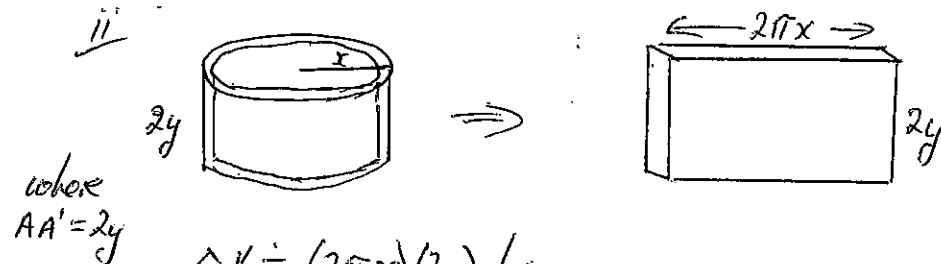
Question 2

b i

$$\begin{aligned} (x-2)^2 + y^2 &= 1 \\ y^2 &= 1 - (x-2)^2 \\ y &= \pm \sqrt{1 - (x-2)^2} \end{aligned}$$

Let 2 points A and A' on the circle are
A(x, $\sqrt{1 - (x-2)^2}$) and A'(x, $-\sqrt{1 - (x-2)^2}$) ✓
 $\therefore AA' = \sqrt{1 - (x-2)^2} - (-\sqrt{1 - (x-2)^2})$
 $= 2\sqrt{1 - (x-2)^2}$ ✓

ii



$$\begin{aligned} \Delta V &\equiv (2\pi x)(2y) dx \\ &= 2\pi x (2\sqrt{1 - (x-2)^2}) dx \\ &= 4\pi x \sqrt{1 - (x-2)^2} dx \end{aligned}$$

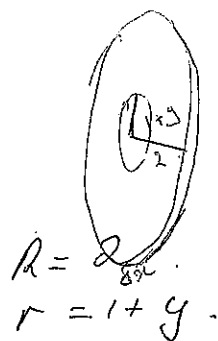
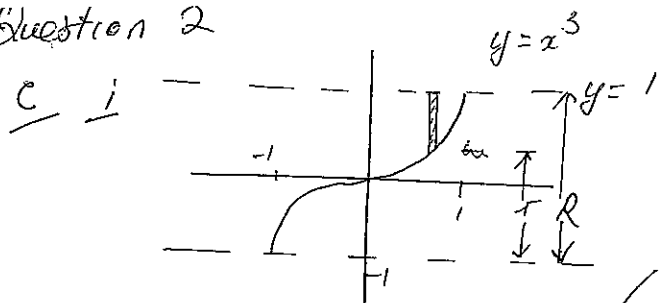
iii

$$\begin{aligned} \text{Volume} &= \lim_{dx \rightarrow 0} \sum_{i=1}^3 \Delta V \\ &= \int_1^3 4\pi x \sqrt{1 - (x-2)^2} dx \end{aligned}$$

Let $u = x-2$, $\therefore du = dx$, $x = u+2$, $x=1$ $u=-1$, $x=3$ $u=1$

$$\begin{aligned} &= 4\pi \int_{-1}^1 (u+2) \sqrt{1-u^2} du \\ &= 4\pi \left[\int_{-1}^1 u \sqrt{1-u^2} du + 2 \int_{-1}^1 \sqrt{1-u^2} du \right] \\ &= 4\pi \left[0 + \pi \right] \quad \text{odd function} \quad \text{semi circle} \\ &= 4\pi^2 \end{aligned}$$

Question 2



$$\begin{aligned}\Delta V &= \pi (R^2 - r^2) dx \\ &= \pi [4 - (1 + y)^2] dx \quad \checkmark \\ &= \pi [4 - (1 + 2y + y^2)] dx \\ &= \pi [3 - 2y - y^2] dx \\ &= \pi [3 - 2x^3 - x^6] dx \quad \checkmark\end{aligned}$$

ii

$$\begin{aligned}\text{Volume} &= \lim_{dx \rightarrow 0} \sum_{x=0}^1 \pi (3 - 2x^3 - x^6) dx \quad \checkmark \\ &= \pi \int_0^1 (3 - 2x^3 - x^6) dx \quad \checkmark \\ &= \pi \left[3x - \frac{2x^4}{4} - \frac{x^7}{7} \right]_0^1 \\ &= \pi \left[3 - \frac{1}{2} - \frac{1}{7} - 0 \right] \\ &= \frac{33\pi}{14} \text{ units}^3 \quad \checkmark\end{aligned}$$